

JSS 2 MATHEMATICS – FIRST TERM

TOPIC 3: LCM AND HCF

1. Highest Common Factor (HCF)

HCF (or GCD) of two or more numbers is the largest number that divides each of them exactly. Method: write prime factors; take the lowest power of each common prime.

Example: HCF of 24 and 36. $24 = 2^3 \times 3$; $36 = 2^2 \times 3^2$. $\text{HCF} = 2^2 \times 3 = 12$.

2. Lowest Common Multiple (LCM)

LCM of two or more numbers is the smallest number that is a multiple of each of them. Method: take the highest power of every prime that appears.

Example: LCM of 24 and 36 $= 2^3 \times 3^2 = 8 \times 9 = 72$.

3. Squares and Square Roots

A perfect square is the product of a number by itself (e.g. $16 = 4^2$). Square root of a number is the value that, when multiplied by itself, gives the number ($\sqrt{16} = 4$). Use prime factors to find square roots of perfect squares.

4. Application in Quantitative Reasoning

LCM and HCF appear in problems about cycles (e.g. when two events coincide), sharing items, and simplifying ratios.

5. Summary

HCF uses the lowest powers of common primes; LCM uses the highest powers of all primes. Squares and square roots are linked to prime factors of perfect squares.

End of Topic 3